

Qualification Pack



Computer Hardware and Networking Associate

QP Code: HAR/ITS/Q0401

Version: 1.0

NSQF Level: 4

Haryana State Electronics Development Corporation Limited || SCO 111-113, Sector 17 B
Chandigarh || email:hartron.vikaspanwar@gmail.com



Qualification Pack

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Qualification Pack

HAR/ITS/Q0401: Computer Hardware and Networking Associate

Brief Job Description

The qualification provides technical knowledge of computer hardware and networking, including system assembly, troubleshooting, LAN/WAN setup, IP addressing, and network security, preparing learners for entry-level IT roles.

Personal Attributes

A person should possess strong problem-solving skills, attention to detail, and technical aptitude. Good communication and adaptability to evolving technologies are also essential.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HAR/ITS/N0403: Fundamentals of IT](#)
2. [HAR/ITS/N0404: Connecting to the World](#)
3. [HAR/ITS/N0401: Basic Electronics \(60 hrs\)](#)
4. [HAR/ITS/N0402: Computer Hardware & Peripheral \(90 hrs\)](#)
5. [HAR/ITS/N0504: Operating System\(45 hrs\)](#)
6. [HAR/ITS/N0405: Network Essentials & Management](#)
7. [HAR/ITS/N0406: Server Installation & Administration](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	IT- Services
Occupation	Computer Hardware and Networking Associate
Country	India
NSQF Level	4



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Credits	19
Aligned to NCO/ISCO/ISIC Code	3513
Minimum Educational Qualification & Experience	12th grade Pass
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	23/06/2026
NSQC Approval Date	23/06/2023
Version	1.0
Reference code on NQR	NCVET-QG-04-IT-00629-2023-V2-HART
NQR Version	2

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HAR/ITS/N0403: Fundamentals of IT

Description

Provides foundational knowledge of Information Technology, covering the basics of computer hardware, software, operating systems, and office automation tools.

Scope

The scope covers the following :

- Equips students with the essential IT skills for using computer systems and office applications efficiently in personal, academic, or professional settings.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define computer and explain its characteristics, applications, advantages & disadvantages
- PC2.** Classify computer based on their generation, & size
- PC3.** Define various components of computer including memory and its types
- PC4.** Define and use file, folders, various icons & Window Accessories options.
- PC5.** Create, edit, and format documents in Microsoft Word, using features such as text formatting, graphics, tables, and mail merge.
- PC6.** Work with Microsoft Excel to manipulate data by applying functions & formulas and insert charts and graphs.
- PC7.** Present information visually and effectively in slides using animations, color schemes and templates

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Objective & goals of organization and scope of work assigned
- KU2.** Basic concepts of a computer and functions of its components.
- KU3.** The working in Windows environment , using essential softwares, utilities and file management system
- KU4.** Core functionalities of Microsoft Office tools (Word, Excel, and PowerPoint) to create and manipulate documents, spreadsheets, and presentations effectively.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Listen actively and communicate with other team members orally and in writing.
- GS2.** Seek inputs and suggestions from team members, client and industry experts.



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- GS3.** Handle multiple tasks concurrently by prioritizing them to meet the deadlines and ensuring accuracy in tasks
- GS4.** Using digital tools for teamwork and file-sharing

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	11	15	5	5
PC1. Define computer and explain its characteristics, applications, advantages & disadvantages	-	-	-	-
PC2. Classify computer based on their generation, & size	-	-	-	-
PC3. Define various components of computer including memory and its types	-	-	-	-
PC4. Define and use file, folders, various icons & Window Accessories options.	-	-	-	-
PC5. Create, edit, and format documents in Microsoft Word, using features such as text formatting, graphics, tables, and mail merge.	-	-	-	-
PC6. Work with Microsoft Excel to manipulate data by applying functions & formulas and insert charts and graphs.	-	-	-	-
PC7. Present information visually and effectively in slides using animations, color schemes and templates	-	-	-	-
NOS Total	11	15	5	5



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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0403
NOS Name	Fundamentals of IT
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Hardware and Networking Associate
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

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HAR/ITS/N0404: Connecting to the World

Description

Equips students with essential skills to navigate the digital world efficiently, communicate effectively through e-mail, and introduction to cloud storage platforms.

Scope

The scope covers the following :

- Prepares students to navigate online services, manage digital communication effectively for personal and professional use.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define Internet, its advantages, disadvantages and applications
- PC2.** Explain the terms Domain Name Server, Web Browser, search Engine, Uniform Resource Locator, e-mail etc.
- PC3.** Navigate through internet , searching, downloading and uploadin file/data using various online services.
- PC4.** Creates and manages e-mail accounts, composing & receiving messages and files throgh e-mail .
- PC5.** Define e-governance services and their types of interactions.
- PC6.** Define Cloud, Cloud storage and types

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational objectives and goals and Scope of work.
- KU2.** The fundamental concepts of the internet , its uses for various purposes and potential security risks.
- KU3.** Navigating websites, Using e-mail services , organising and managing e-mails
- KU4.** The basic concepts of cloud services and using cloud storage (public, private, hybrid, community)
- KU5.** How e-mail communication and cloud storage help businesses and organizations by improving productivity, teamwork, and managing data effectively.
- KU6.** The significance of e-governance and its application in modern digital and governmental systems, helping users to navigate public services online.

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** Use online resources, web browsers, and digital tools effectively to find information and solve tasks.
- GS2.** Effectively manages e-mail communication and organizes online data to improve workflow and productivity.
- GS3.** Understanding email etiquette, including appropriate tone and formatting
- GS4.** Identify and solve problems related to internet connectivity or e-mail issues
- GS5.** The importance of Safe and secure net surfing and and accessing & protecting sensitive data through cloud services

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	11	15	5	5
PC1. Define Internet, its advantages, disadvantages and applications	-	-	-	-
PC2. Explain the terms Domain Name Server, Web Browser, search Engine, Uniform Resource Locator, e-mail etc.	-	-	-	-
PC3. Navigate through internet , searching, downloading and uploadin file/data using various online services.	-	-	-	-
PC4. Creates and manages e-mail accounts, composing & receiving messages and files throgh e-,mail .	-	-	-	-
PC5. Define e-governance services and their types of interactions.	-	-	-	-
PC6. Define Cloud, Cloud storage and types	-	-	-	-
NOS Total	11	15	5	5



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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0404
NOS Name	Connecting to the World
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Hardware and Networking Associate
NSQF Level	4
Credits	1.5
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

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HAR/ITS/N0401: Basic Electronics (60 hrs)

Description

Provides foundational knowledge and skills to perform basic tasks involving electronic circuits and components of computer

Scope

The scope covers the following :

- Prepare learners to pursue careers in electronics repair and manufacturing of computer devices.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define current, voltage & Resistance and explain the relationship between them
- PC2.** Describe the functions of resistors, capacitors, and inductors in an electronic circuit
- PC3.** Identify and describe various active and passive components.
- PC4.** Describe Integrated Circuits(IC) and Semiconductor memory
- PC5.** Explain various types of power supplies used in electronics
- PC6.** Define various Number system and perform conversions between decimal and other number systems, and vice versa.
- PC7.** Draw and explain various logic gates and combinational logic circuits such as half-adder and half-subtractor with their truth tables.
- PC8.** Define and differentiate between encoders and decoders, Multiplexers and De-multiplexers. Explain with diagrams and truth tables for each.
- PC9.** Draw and explain various types of Flip Flops, Latches, Register and Counters.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The objective and goals of organization and scope of work assigned
- KU2.** The basic principles of electricity, including current, voltage, and resistance.
- KU3.** The working principles of active and passive electronic components.
- KU4.** The use of tools like multimeters, oscilloscopes, and soldering irons for designing, testing, and troubleshooting analog and digital circuits.
- KU5.** Workplace safety practices, precautions and equipment handling.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to convey technical concepts clearly and effectively, both orally and in writing



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- GS2.** Analyze and diagnose issues in electronic circuits using logical reasoning and systematic methods.
- GS3.** Stay updated with the latest advancements in electronics technology.
- GS4.** Work effectively in teams to assemble, test, and troubleshoot electronic systems.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	13	-	10	10
PC1. Define current, voltage & Resistance and explain the relationship between them	-	-	-	-
PC2. Describe the functions of resistors, capacitors, and inductors in an electronic circuit	-	-	-	-
PC3. Identify and describe various active and passive components.	-	-	-	-
PC4. Describe Integrated Circuits(IC) and Semiconductor memory	-	-	-	-
PC5. Explain various types of power supplies used in electronics	-	-	-	-
PC6. Define various Number system and perform conversions between decimal and other number systems, and vice versa.	-	-	-	-
PC7. Draw and explain various logic gates and combinational logic circuits such as half-adder and half-subtractor with their truth tables.	-	-	-	-
PC8. Define and differentiate between encoders and decoders, Multiplexers and De-multiplexers. Explain with diagrams and truth tables for each.	-	-	-	-
PC9. Draw and explain various types of Flip Flops, Latches, Register and Counters.	-	-	-	-
NOS Total	13	-	10	10



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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0401
NOS Name	Basic Electronics (60 hrs)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Hardware and Networking Associate
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

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HAR/ITS/N0402: Computer Hardware & Peripheral (90 hrs)

Description

Equips learners with the skills required for computer hardware maintenance, assembly, and troubleshooting.

Scope

The scope covers the following :

- Provides essential knowledge and practical skills in assembling, disassembling, and troubleshooting computer systems and peripherals and prepares learners for careers in IT support, hardware maintenance, and system management.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define and explain the functions of various internal and external computer components.
- PC2.** Define process for Assembling and Disassembling the computer.
- PC3.** Recognize and describe the use of different internal ports, cables, and connectors used for system integration and communication.
- PC4.** Explain the use of Modem, Digital Camera, Pen drives, SCSI interface device, Web Camera, CCTV Camera, Projector
- PC5.** Describe various types of monitors and their differences in terms of display technology and application.
- PC6.** Describe the different types of printers and scanners, and explain their functions in the computer ecosystem.
- PC7.** Identify and troubleshoot common hardware and software issues

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The objective & goals of organization and scope of work assigned.
- KU2.** The components of a computer system and how they interact with each other to form a functional unit.
- KU3.** The assembly and disassembly processes, including the installation of components and internal wiring and configuring a computer system.
- KU4.** How to diagnose and resolve hardware-related issues, including malfunctioning devices, slow performance, and connectivity problems.
- KU5.** The role and importance of computer hardware in different organizational environments,
- KU6.** How to apply workplace safety standards during the assembly, disassembly, and troubleshooting of computer hardware.



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Make informed decisions when selecting hardware components or troubleshooting system issues, ensuring solutions that are both efficient and cost-effective.
- GS2.** Effectively use tools and equipment for assembling, disassembling, and troubleshooting computers .
- GS3.** Apply logical thinking and systematic troubleshooting to resolve technical issues.
- GS4.** Understand client needs and deliver solutions that meet performance, cost, and reliability requirements, ensuring customer satisfaction.
- GS5.** Prioritize tasks effectively, especially during troubleshooting scenarios, to meet deadlines.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	-	10	10
PC1. Define and explain the functions of various internal and external computer components.	-	-	-	-
PC2. Define process for Assembling and Disassembling the computer.	-	-	-	-
PC3. Recognize and describe the use of different internal ports, cables, and connectors used for system integration and communication.	-	-	-	-
PC4. Explain the use of Modem, Digital Camera, Pen drives, SCSI interface device, Web Camera, CCTV Camera, Projector	-	-	-	-
PC5. Describe various types of monitors and their differences in terms of display technology and application.	-	-	-	-
PC6. Describe the different types of printers and scanners, and explain their functions in the computer ecosystem.	-	-	-	-
PC7. Identify and troubleshoot common hardware and software issues	-	-	-	-
NOS Total	20	-	10	10

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0402
NOS Name	Computer Hardware & Peripheral (90 hrs)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Hardware and Networking Associate
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQF Clearance Date	23/06/2023

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HAR/ITS/N0504: Operating System(45 hrs)

Description

Provides essential knowledge of operating system concepts, installation, and maintenance.

Scope

The scope covers the following :

- Provides knowledge and skills in installing, configuring, and troubleshooting operating systems and virus protection techniques, ensuring system security and maintenance in professional environments.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the role of an operating system and its types
- PC2.** Define and differentiate System files(FAT and NTFS) and explain the relation between System File and operating System
- PC3.** Install and set operating systems with essential system settings and required application softwares.
- PC4.** Effectively troubleshoot operating system issues, including system crashes, software malfunctions, and security threats, by using appropriate tools
- PC5.** Convert FAT Partition to NTFS
- PC6.** Explain Computer viruses and its types (Malware, Trojan, Worm, Spyware, Adware).
- PC7.** Identify common signs of virus infections, such as window popups, self-executing programs, unexpected logouts, and device crashes.
- PC8.** Apply virus protection measures, including the use of antivirus software, updating the operating system, and using privacy settings in browsers.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The objective & goals of Organization and scope of work assigned.
- KU2.** Fundamental concepts of operating systems, its functions and system files
- KU3.** The principles of virus protection, types of virus and their impact on system security.
- KU4.** The practical skills in installing and configuring operating systems, application software, and system files on different platforms
- KU5.** how to troubleshoot and resolve common operating system issues, from installation errors to system crashes.
- KU6.** The role of operating systems in the day-to-day functioning of an organization's IT infrastructure, from personal computers to server environments.

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Plan and organize tasks for installing, configuring, and maintaining operating systems, ensuring that processes are completed efficiently and within set timelines.
- GS2.** Identifying and resolving OS-related issues effectively.
- GS3.** Effectively present technical information and discuss system security measures in team meetings or with clients.
- GS4.** Deliver user-friendly solutions for system configuration, software installation, and troubleshooting, ensuring that end users or clients experience minimal disruption to their workflow.
- GS5.** Follow organizational policies for OS installation, software updates, and virus protection.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	16	15	8	8
PC1. Explain the role of an operating system and its types	-	-	-	-
PC2. Define and differentiate System files(FAT and NTFS) and explain the relation between System File and operating System	-	-	-	-
PC3. Install and set operating systems with essential system settings and required application softwares.	-	-	-	-
PC4. Effectively troubleshoot operating system issues, including system crashes, software malfunctions, and security threats, by using appropriate tools	-	-	-	-
PC5. Convert FAT Partition to NTFS	-	-	-	-
PC6. Explain Computer viruses and its types (Malware, Trojan, Worm, Spyware, Adware).	-	-	-	-
PC7. Identify common signs of virus infections, such as window popups, self-executing programs, unexpected logouts, and device crashes.	-	-	-	-
PC8. Apply virus protection measures, including the use of antivirus software, updating the operating system, and using privacy settings in browsers.	-	-	-	-
NOS Total	16	15	8	8

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0504
NOS Name	Operating System(45 hrs)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Hardware and Networking Associate
NSQF Level	4
Credits	1.5
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQF Clearance Date	23/06/2023

Qualification Pack

HAR/ITS/N0405: Network Essentials & Management

Description

Provides a comprehensive understanding of networking fundamentals, hardware, IP addressing, wireless technologies, troubleshooting, and security

Scope

The scope covers the following :

- Prepares learners for careers in IT and networking by equipping them with skills to design, troubleshoot, and secure networks.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define and understand the basic concepts of network and its types
- PC2.** Describe computer network architecture and differentiate between them.
- PC3.** Define and Differentiate between LAN, WAN and MAN.
- PC4.** Explain the different types of topologies and Differentiate between them.
- PC5.** Describe transmission media and its types.
- PC6.** Explain OSI and TCP/IP models and differentiate between them
- PC7.** Define protocol and explain different types of internet protocols
- PC8.** Identify and explain the purpose of devices like routers, switches, hubs, and gateways.
- PC9.** Define and configure IP Address
- PC10.** Explain Classes of IP addresses and range of each class
- PC11.** Explain the functionality of security components like firewalls, VPNs, and intrusion prevention systems.
- PC12.** Describe and use tools and methods to identify and resolve network problems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The objective & goals of Organization and scope of work assigned.
- KU2.** Fundamental concepts of networking and its role in communication.
- KU3.** Use of routers, switches, access points, cables and their practical applications.
- KU4.** Networking Protocols and topologies and their suitability for organizational needs.
- KU5.** Applications of Intranets and Extranets in business for secure communication and collaboration.
- KU6.** Importance of protocols in ensuring efficient and secure organizational data exchange.
- KU7.** IP addressing, subnetting, and configuring devices for efficient communication within networks.



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Collaborate effectively with teams and departments to ensure seamless network operations and support.
- GS2.** Provide user-friendly solutions to meet organizational and client requirements.
- GS3.** Stay adaptable to handle evolving networking challenges and technological advancements.
- GS4.** Plan and organize tasks like network setup, maintenance schedules, and upgrades to ensure optimal performance.
- GS5.** Identify and analyze common networking issues, develop logical troubleshooting steps, and propose effective solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	15	12	10
PC1. Define and understand the basic concepts of network and its types	-	-	-	-
PC2. Describe computer network architecture and differentiate between them.	-	-	-	-
PC3. Define and Differentiate between LAN, WAN and MAN.	-	-	-	-
PC4. Explain the different types of topologies and Differentiate between them.	-	-	-	-
PC5. Describe transmission media and its types.	-	-	-	-
PC6. Explain OSI and TCP/IP models and differentiate between them	-	-	-	-
PC7. Define protocol and explain different types of internet protocols	-	-	-	-
PC8. Identify and explain the purpose of devices like routers, switches, hubs, and gateways.	-	-	-	-
PC9. Define and configure IP Address	-	-	-	-
PC10. Explain Classes of IP addresses and range of each class	-	-	-	-
PC11. Explain the functionality of security components like firewalls, VPNs, and intrusion prevention systems.	-	-	-	-
PC12. Describe and use tools and methods to identify and resolve network problems.	-	-	-	-
NOS Total	20	15	12	10



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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0405
NOS Name	Network Essentials & Management
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Hardware and Networking Associate
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

HAR/ITS/N0406: Server Installation & Administration

Description

Provides comprehensive knowledge and hands-on experience in server installation, configuration, and administration.

Scope

The scope covers the following :

- Equips learners with the skills to install, configure, and manage server environments, ensuring seamless operation of enterprise IT infrastructures.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define server is, its types, and the system requirements for installation.
- PC2.** Demonstrate the installation of Windows Server, Linux, or Unix Server.
- PC3.** Configure essential server roles such as Active Directory, DNS, DHCP, IIS, File Server, and Print Server.
- PC4.** Create and manage user accounts, groups, and permissions within Active Directory.
- PC5.** Install & Configure DNS, DHCP, IIS Web Server, File Server, Print Server
- PC6.** Use Group Policy management tools to manage security policies.
- PC7.** Back up, restore, import, and copy Group Policy Objects through the Group Policy

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The objective & goals of Organization and scope of work assigned
- KU2.** The fundamentals of servers, types of servers, and their roles in IT environments.
- KU3.** The concepts of Active Directory, DNS, DHCP, IIS, and storage technologies like DAS, NAS, SAN, and RAID.
- KU4.** The role of servers in enterprise environments for hosting applications, managing users, and supporting business operations.
- KU5.** Managing server environments efficiently, ensuring minimal downtime and optimized performance for organizational needs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate effectively and Collaborate with IT teams to resolve server-related issues.



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- GS2.** Diagnose server errors, analyze root causes, and implement effective solutions to minimize downtime.
- GS3.** Make informed decisions about server hardware, operating systems, and configurations based on organizational needs.
- GS4.** To meet the needs of users and organizational objectives by providing server solutions

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	13	-	8	8
PC1. Define server is, its types, and the system requirements for installation.	-	-	-	-
PC2. Demonstrate the installation of Windows Server, Linux, or Unix Server.	-	-	-	-
PC3. Configure essential server roles such as Active Directory, DNS, DHCP, IIS, File Server, and Print Server.	-	-	-	-
PC4. Create and manage user accounts, groups, and permissions within Active Directory.	-	-	-	-
PC5. Install & Configure DNS, DHCP, IIS Web Server, File Server, Print Server	-	-	-	-
PC6. Use Group Policy management tools to manage security policies.	-	-	-	-
PC7. Back up, restore, import, and copy Group Policy Objects through the Group Policy	-	-	-	-
NOS Total	13	-	8	8

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0406
NOS Name	Server Installation & Administration
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Hardware and Networking Associate
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment is done online after completion of course. Assessment comprises of objective, subjective and practical questions

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HAR/ITS/N0403.Fundamentals of IT	11	15	5	5	36	12
HAR/ITS/N0404.Connecting to the World	11	15	5	5	36	12
HAR/ITS/N0401.Basic Electronics (60 hrs)	13	-	10	10	33	11
HAR/ITS/N0402.Computer Hardware & Peripheral (90 hrs)	20	-	10	10	40	13
HAR/ITS/N0504.Operating System(45 hrs)	16	15	8	8	47	16
HAR/ITS/N0405.Network Essentials & Management	20	15	12	10	57	19
HAR/ITS/N0406.Server Installation & Administration	13	-	8	8	29	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	7
Total	124	90	58	56	328	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.